

Lab ASSIGNMENT – 6.5

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BATH – 29

Task Description #1 (AI-Based Code Completion for Conditional Eligibility Check)

Task: Use an AI tool to generate eligibility logic.

Prompt:

“Generate Python code to check voting eligibility based on age and citizenship.”

Expected Output:

- AI-generated conditional logic.
- Correct eligibility decisions.
- Explanation of conditions.

CODE:

```
def check_voting_eligibility(age, is_citizen):
```

```
    if age >= 18 and is_citizen:
```

```
        return "Eligible to vote"
```

```
    else:
```

```
        return "Not eligible to vote"
```

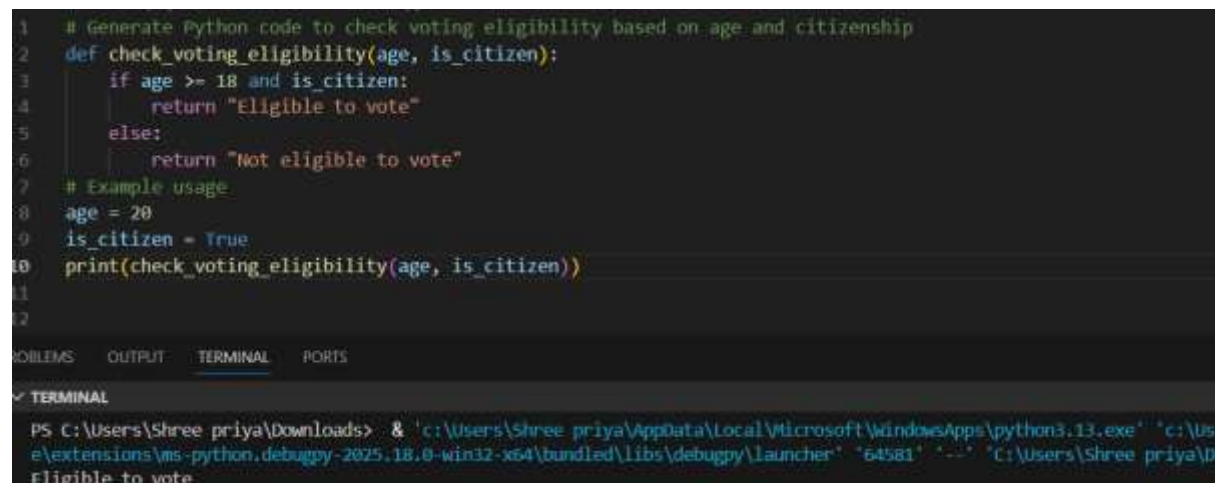
```
# Example usage
```

```
age = 20
```

```
is_citizen = True
```

```
print(check_voting_eligibility(age, is_citizen))
```

OUTPUT: Eligible to vote



```
1 # Generate Python code to check voting eligibility based on age and citizenship
2 def check_voting_eligibility(age, is_citizen):
3     if age >= 18 and is_citizen:
4         return "Eligible to vote"
5     else:
6         return "Not eligible to vote"
7 # Example usage
8 age = 20
9 is_citizen = True
10 print(check_voting_eligibility(age, is_citizen))
11
12
```

PROBLEMS OUTPUT TERMINAL PORTS

✓ TERMINAL

```
PS C:\Users\Shree priya\Downloads> & 'c:\Users\Shree priya\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Us
e\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '64581' '...' 'C:\Users\Shree priya\D
Eligible to vote
```

Task Description #2(AI-Based Code Completion for Loop-Based String Processing)

Task: Use an AI tool to process strings using loops.

Prompt:

“Generate Python code to count vowels and consonants in a string using a loop.”

Expected Output:

- AI-generated string processing logic.
- Correct counts.
- Output verification

CODE:

```
def count_vowels_and_consonants(input_string):  
    vowels = "aeiouAEIOU"  
    vowel_count = 0  
    consonant_count = 0  
    for char in input_string:  
        if char.isalpha():  
            if char in vowels:  
                vowel_count += 1  
            else:  
                consonant_count += 1  
    return vowel_count, consonant_count  
  
# Example usage  
input_string = "Hello World"  
vowels, consonants = count_vowels_and_consonants(input_string)  
print(f"Vowels: {vowels}, Consonants: {consonants}")
```

OUTPUT: Vowels: 3, Consonants: 7

```
def count_vowels_and_consonants(input_string):
    vowels = "aeiouAEIOU"
    vowel_count = 0
    consonant_count = 0

    for char in input_string:
        if char.isalpha():
            if char in vowels:
                vowel_count += 1
            else:
                consonant_count += 1

    return vowel_count, consonant_count

# Example usage
input_string = "Hello World"
vowels, consonants = count_vowels_and_consonants(input_string)
print(f"Vowels: {vowels}, Consonants: {consonants}")
```

TERMS OUTPUT TERMINAL PORTS

TERMINAL

Vowels: 3, Consonants: 7

Task Description #3 (AI-Assisted Code Completion Reflection Task)

Task: Use an AI tool to generate a complete program using classes, loops, and conditionals.

Prompt:

“Generate a Python program for a library management system using classes, loops, and conditional statements.”

Expected Output:

- Complete AI-generated program.
- Review of AI suggestions quality.
- Short reflection on AI-assisted coding experience.

CODE:

class Library:

def __init__(self):

self.books = ["Python", "Java"]

lib = Library()

while True:

print("\n1.Show Books 2.Issue Book 3.Exit")

ch = input("Choice: ")

if ch == "1":

for b in lib.books:

```

        print(b)
elif ch == "2":
    book = input("Book name: ")
    if book in lib.books:
        lib.books.remove(book)
        print("Book issued")
    else:
        print("Not available")
elif ch == "3":
    print("Bye")
    break
else:
    print("Wrong choice")
# Example usage
books = Library()
books.books.append("C++")
books.books.append("JavaScript")
print("Available books:")
for book in books.books:
    print(book)

```

OUTPUT : 1.Show Books 2.Issue Book 3.Exit

Choice: 2

Book name: Python

Book issued

1.Show Books 2.Issue Book 3.Exit

Choice: 1

Java

1.Show Books 2.Issue Book 3.Exit

Choice: 3

Bye

Available books:

Python

Java

C++

JavaScript

```
# Generate a Python program for a library management system using classes, loops, and conditions
class Library:
    def __init__(self):
        self.books = ["Python", "Java"]
lib = Library()
while True:
    print("\n1.Show Books  2.Issue Book  3.Exit")
    ch = input("Choice: ")
    if ch == "1":
        for b in lib.books:
            print(b)
    elif ch == "2":
        book = input("Book name: ")
        if book in lib.books:
            lib.books.remove(book)
            print("Book issued")
        else:
            print("Not available")
    elif ch == "3":
        print("Bye")
        break
    else:
        print("Wrong choice")
# Example usage
books = Library()
books.books.append("C++")
books.books.append("JavaScript")
print("Available books:")
for book in books.books:
    print(book)
```

IDEA OUTPUT TERMINAL PORTS

TERMINAL

Choice: 1
Java

1.Show Books 2.Issue Book 3.Exit
Choice: 2
Book name: Python Java
Not available

1.Show Books 2.Issue Book 3.Exit
Choice: 3
Bye
Available books:
Python
Java
C++
JavaScript

Task Description #4 (AI-Assisted Code Completion for Class- Based Attendance System)

Task: Use an AI tool to generate an attendance management class.

Prompt: "Generate a Python class to mark and display student attendance using loops."

Expected Output:

- AI-generated attendance logic.
- Correct display of attendance.
- Test cases.

CODE:

class StudentAttendance:

def __init__(self):

self.attendance = {}

```
def mark_attendance(self, student_name):  
    self.attendance[student_name] = "Present"
```

```
def display_attendance(self):  
    for student, status in self.attendance.items():  
        print(f"{student}: {status}")
```

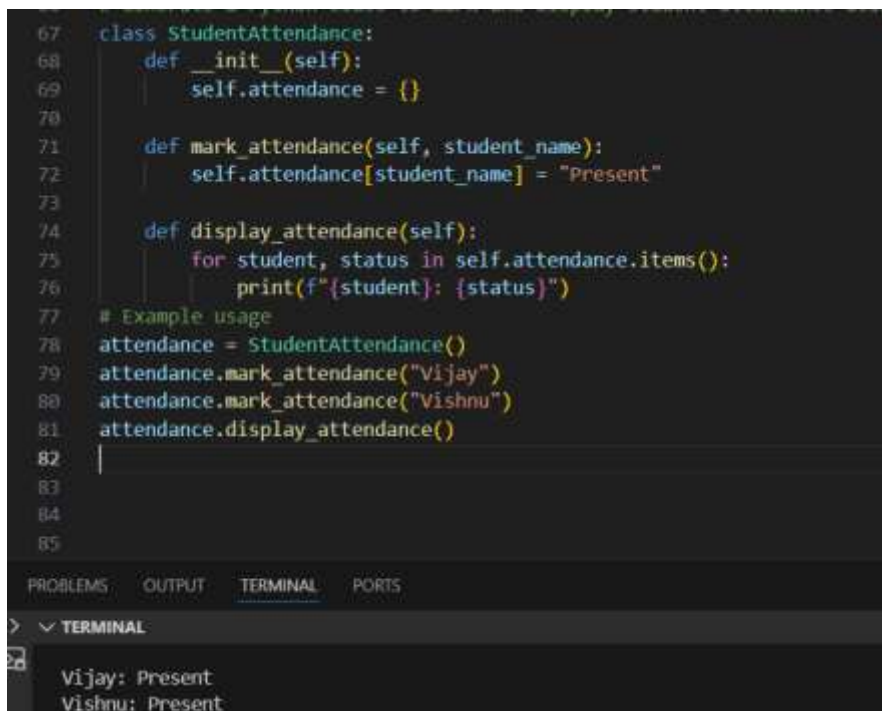
Example usage

```
attendance = StudentAttendance()  
attendance.mark_attendance("Vijay")  
attendance.mark_attendance("Vishnu")  
attendance.display_attendance()
```

OUTPUT:

Vijay: Present

Vishnu: Present



```
67 class StudentAttendance:  
68     def __init__(self):  
69         self.attendance = {}  
70  
71     def mark_attendance(self, student_name):  
72         self.attendance[student_name] = "Present"  
73  
74     def display_attendance(self):  
75         for student, status in self.attendance.items():  
76             print(f"{student}: {status}")  
77  
78 # Example usage  
79 attendance = StudentAttendance()  
80 attendance.mark_attendance("Vijay")  
81 attendance.mark_attendance("Vishnu")  
82 attendance.display_attendance()  
83  
84  
85
```

PROBLEMS OUTPUT TERMINAL PORTS

> ▾ TERMINAL

```
Vijay: Present  
Vishnu: Present
```

Task Description #5 (AI-Based Code Completion for Conditional Menu Navigation)

Task: Use an AI tool to complete a navigation menu.

Prompt: "Generate a Python program using loops and conditionals to simulate an ATM menu."

Expected Output:

- AI-generated menu logic.
- Correct option handling.
- Output verification

CODE:

```
bal = 1000

while True:

    ch = input("\n1.Balance 2.Deposit 3.Withdraw 4.Exit : ")

    if ch == "1":

        print("Balance:", bal)

    elif ch == "2":

        bal += int(input("Amount: "))

    elif ch == "3":

        a = int(input("Amount: "))

        if a <= bal: bal -= a

        else: print("No money")

    elif ch == "4":

        break
```

OUTPUT:

1.Balance 2.Deposit 3.Withdraw 4.Exit : 1

Balance: 1000

1.Balance 2.Deposit 3.Withdraw 4.Exit : 2

Amount: 1000000

1.Balance 2.Deposit 3.Withdraw 4.Exit : 3

Amount: 5000

1.Balance 2.Deposit 3.Withdraw 4.Exit : 4

```

87 bal = 1000
88 while True:
89     ch = input("\n1.Balance 2.Deposit 3.Withdraw 4.Exit : ")
90
91     if ch == "1":
92         print("Balance:", bal)
93
94     elif ch == "2":
95         bal += int(input("Amount: "))
96
97     elif ch == "3":
98         a = int(input("Amount: "))
99         if a <= bal: bal -= a
100        else: print("No money")
101
102     elif ch == "4":
103         break

```

PROBLEMS OUTPUT TERMINAL PORTS

▼ TERMINAL

```

JavaScript
Vijay: Present
Vishnu: Present

```

```

1.Balance 2.Deposit 3.Withdraw 4.Exit : 1
Balance: 1000

```

```

1.Balance 2.Deposit 3.Withdraw 4.Exit : 2
Amount: 1000000

```

```

1.Balance 2.Deposit 3.Withdraw 4.Exit : 3
Amount: 5000

```

```

1.Balance 2.Deposit 3.Withdraw 4.Exit : 4
PS C:\Users\Shomendra\Downloads>

```